

EFFICIENT AND ROBUST FOOT PLACEMENT CONTROL ENABLED BY REINFORCEMENT LEARNING AND PASSIVE DYNAMICS

Qi Tao, Yiyu Liu, Amir Degani, Hao Tang, Ray Kong, Mingyi Liu

Abstract - Inspired by humans, bipedal walking robots have been extensively studied theoretically and experimentally in a broad range of literature. ATLAS, ASIMO, and Cassie have demonstrated the fruitful outcomes of this field. However, to date, walking robots still falls far behind humans in most performance indices, such as agility, smoothness, robustness, and efficiency. Foot placement is critical to bipedal walking and the control is challenging due to the underactuated and nonlinear dynamics. To tackle this challenge, we propose an efficient and robust foot placement control algorithm based on reinforcement learning and passive dynamics. With a specially designed action setup in the training process, the robot can control the swing leg and achieve the predefined target landing position robustly with minimum energy. The robot can also absorb large random disturbances and still reach the predefined target landing position. Even with a changing target position, the robot can still adjust its controlling strategy and eventually reach the predefined target position. A simple prototype was built to validate the concept. The theoretical results and simulations are validated by experiment and the foot placement control has proved to be efficient and robust.

Index Terms—Bipedal walking, passive dynamics, reinforcement learning, efficiency

I. INTRODUCTION

Through millions of years of evolution, human motion has been optimized to be efficient and robust. By observing human walking, it is hard to ignore the beauty of the planning, control, and execution. In addition, the not-so-obvious concepts behind the curtain were also studied thoroughly in literature, e.g., passive dynamics, optimal control, and reinforcement learning. Human locomotion requires precise coordination between the biomechanics of the bipedal gait cycle and the eye movements that gather the information needed to guide foot placement [1]. For centuries, researchers have been inspired and motivated to study the dynamics and control in various domains, trying to design robots to catch-up with human abilities. Kinesiology, neurology, and motor control studies have improved our understanding of intricate human motions and underlying motor behaviors [2]. At the same time, the booming development in the field of reinforcement learning has

also provided new tools to study human walking and design robots to catch up with human performance. Bipedal robots have gradually evolved from science fiction to reality.

Passive dynamics refers to using the natural dynamics response of the system to achieve controls, first proposed by McGeer [3], and subsequently studied by multiple scientists. This phenomenon is observed in many animals walking and even running [4]. Because the motion is a result of the natural dynamics of the system, rather than the closed-loop trajectory following [5], the actuators are at rest for most of the time. As a result, the system can be more efficient. The high energy efficiency of human locomotion is the result of our ability to exploit the passive dynamics of bipedal gait, rather than active muscle control. Inspired by the lower-energy-cost walking of humans, bipedal robots that built on passive dynamics were invented and showcased the efficiency of passive dynamics [6].

Foot placement is a critical component in walking. In the early 1970s, Waseda University built the robot WABOT that used static stability as the control paradigm [7], and Vukobratovic of Mihajlo Pupin Institute designed the first active exoskeleton [8], by applying the zero-moment point (ZMP) [9]. Although this method has been successfully applied to many robots, such as Honda ASIMO robot [10], HRP series [11] and HUBO [12], the ZMP standard still has serious drawbacks. One simple fact to check is that human walking does not follow the ZMP rules. During human walking, the center of mass (CoM) is outside the base of support for most of the time, which poses a challenge to balance. Nevertheless, most healthy individuals are able to walk without any problems. Humans achieve this through foot placement via hip abductor muscle during the swing phase of gait. To achieve dynamic motion, researchers began to try to find a way to utilize the natural dynamics of the system. The MIT Leg Lab used reduced-order models (such as springs and inverted pendulums) to create stable periodic orbits [13] to implement the running and flipping behaviors of jumping robots [14]. Around 1990, McGeer began to study completely passive walking robots, which eventually led to the creation of passive dynamic walking [3]. This method demonstrates the feasibility of robot motion on underactuated systems and demonstrates the possibility of exploiting under-actuation to improve robot performance. However, the robot's actuation authority is severely limited by the lack of formal methods that exploit the nature of motion dynamics. In early 2000, Grizzle proposed the concept of virtual constraints [15], that is, complete constraints not through physical mechanisms but through control. By

designing these constraints, the concept of hybrid zero dynamics (HZD) [16] was eventually produced, resulting in a guarantee for the generation and stability of periodic orbits [17]. Based on this, the control of dynamic underactuated motion has ushered in rapid progress and subsequently achieved fast running [18][19] and efficient humanoid walking [20][21].

Reinforcement learning is a machine learning (ML) technique that trains agents to make optimal decisions. It was first proposed as an independent field of computer science by psychologists and computer scientists [22]. Reinforcement learning has undergone explosive development in recent years, demonstrated in playing computer games, playing Go [23], as well as robot control [24]. Because the control rules are learned rather than explicitly programmed, reinforcement-learning-based control can deal with nonlinear, piecewise, or even model-free systems. Many researchers are trying to use the booming artificial intelligence in recent years to achieve optimal feedback strategies for robots. Methods such as stochastic gradients, deep reinforcement learning (DRL) have been explored for bipeds, quadrupeds, and hexapods [25][26]. Another obvious advantage of the reinforcement learning is that a control action map that covers the entire state space can be produced such that no matter where the initial state is, the control strategy can lead to the goal.

It is obvious that when a human walks, both reinforcement-learning-based control and passive dynamics are behind the curtain. There is work that apply reinforcement learning and passive dynamics, but they did not consider foot placement [27]. In [28], the authors tried to use reinforcement learning to get the motion trajectory back to one that is defined by passive dynamics. [29] show through reinforcement learning, various goals like maximum speed and minimal cost of transport, and adaptation to unknown situations can be quickly found by simulation model. [30] studied a reinforcement learning scheme designed in particular for Quasi Passive Dynamic Walking (PDW) that is robust against sudden perturbations and variations in the slope gradient. However, almost all of them consider other goals but the foot placement. In human walking, people need to consider the next spot our foot can place on, especially when it comes to uneven terrains, such as stairs and dirt roads.

Up to date, no work has bridged reinforcement learning and passive dynamics in foot placement, while human beings always use reinforcement learning and passive dynamic in our foot placement. The natural question is: why not bridge reinforcement learning and passive dynamics, thus enable robust and efficient bipedal walking?

In this paper, we explore the possibility of bridging reinforcement learning and passive dynamics, enabling robust and efficient foot placement control. A human walking foot placement control problem is used to test this concept. By action profile setup and training process, passive dynamics can

be achieved. We have formulated the problem, built the system model, designed the reinforcement learning strategy, finalized the training, and obtained the action map. The main contributions are:

In an action sequence, only one direction action torque is allowed. Positive actuation and negative action cannot show up in the same control action sequence. Thus, the system is ensured to be passive-based and energy efficient.

Limit the action torque to three options: a fixed positive value, a zero, and a fixed negative value, rather than analog so that the redundancy in action space is reduced. Thus, enable unique optimal solution and improve the training quality.

Train the entire action map and find out the basin of attraction after imposing a control strategy, thus enabling achieving the target goal with large area of initial position and random perturbations.

Train multiple action maps based on different targets and eventually get the action map as a function of the system states, as well as the targets. Thus enable the robot to deal with sudden changes in target positions.

The rest of the paper is organized as follows. In [Section 2](#), the background is introduced. Control system design is built based on reinforcement learning and passive dynamics. The problem formulation training process is described in detail and the action map was obtained in [Section 3](#). [Section 4](#) is simulation result and validation. In [Section 5](#), hardware was built to validate the concept and the simulation result. Discussion and conclusion are drawn and presented in [Section 6](#).

II. BACKGROUND

A. Foot placement and reinforcement learning

Foot placement is essential to human locomotion [31][32]. By utilizing information about upcoming footholds, walkers are able to accurately step on their desired location while efficiently exploiting the natural dynamics of bipedal gait [29]. The study in [33] shows that with two steps of visual look-ahead, the walker provides sufficient information to control the placement of the foot. The human walking model can be simplified by a multiple DOF pendulum [13]. Foot placement's role in the control of walking robots is also considered indispensable [34]. Due to the pendulum dynamics (with sine and cosine terms in the equation of motion), the system is intrinsic nonlinear. The control of the human walking model thus cannot use simple linear control theory. The human walking model is extensively studied in literature, but the performance of walk robots is still not comparable to that of humans.

Reinforcement Learning (RL) is concerned with how to map situations to actions by interacting with the environment and maximizing the cumulative reward received from the environment. This process coincides with the process by which people learn the behavioral pattern of walking, so

reinforcement learning is undoubtedly very suitable for robots to learn to walk.

B. Markov Decision Process

Markov Decision Processes (MDPs) are a core concept in reinforcement learning, used to model decision-making problems, where an agent interacts with an environment to achieve a goal. The goal of MDPs is to find an optimal policy $\pi(a | s)$ that defines the probability of choosing action a_t in state s_t and maximizes the expected cumulative reward R from following that policy starting from any initial state s_t . The expected cumulative reward is usually expressed as:

$$G_t = \sum_{k=0}^{\infty} \gamma^k r_{t+k+1}, 0 \leq \gamma \leq 1,$$

where G_t is the cumulative discounted reward starting from time step t ($t = 0, 1, 2, 3, \dots$) and the discount factor γ is used to measure the importance of future rewards.

C. Actor-Critic

For the continuous state space of real-world problems, it is not feasible to train each state independently, and discretized grid training will lead to the accumulation of errors. Reinforcement learning algorithms based on simple value iteration or value iteration each have the problem of low efficiency or low stability. In order to address the shortcomings of the above two methods, Actor-Critic combines policy gradient and value function. Actor is used to approximate the policy function, which defines the probability distribution of taking each possible action a in a given state s . Critic is used to approximate the value function, which is responsible for evaluating the expected cumulative reward that can be obtained by following the current policy in a given state s , thereby providing feedback to the Actor on the policy performance.

D. Proximal Policy Optimization

Proximal Policy Optimization (PPO) [35] is a reinforcement learning algorithm, meticulously crafted to offer a robust and efficient alternative to the conventional policy gradient method. The essence of the PPO algorithm resides in its commitment to ensuring the stability of the training process and mitigating the risk of overzealous policy updates. This is achieved through the introduction of a pivotal constraint known as "proximal policy optimization," which serves to regulate the disparity between the incumbent and the revised policies, preventing them from diverging excessively during the update phase. The salient attributes of PPO include (Trust Region Optimization), Clipped Surrogate Objective, (Adaptive KL Penalty Coefficient) and Multi-step Updates [36].

III. METHODS

A. Problem Formulation

Foot placement entails a look-ahead mechanism, where the eyes perceive incoming terrain and relay this information to the

brain, enabling it to plan the subsequent foot placement location (see Figure 1A). Consequently, the brain maps out an optimal path for leg swing and coordinates whole-body muscles to ensure precise landing. Even though the coordination involves various muscle groups - from the hands and hips to the ankles - literature posits the ankle and the hip as the most vital factor in driving this process [37]. Therefore, in our streamlined model, the minimal contributions made by the hands, knee, and ankles are disregarded. More actuators mean more complex system. In this current work, for simplicity reasons, only one step foot placement is considered so only one actuator located at the hip is required for passive foot control. As illustrated in Figure 1B, the system is made up of two poles (two legs) connected by one joint (the hip). The goal is to control the placement of the swing leg on the target position which in the example shown in Figure 1A is the top of the next rock.

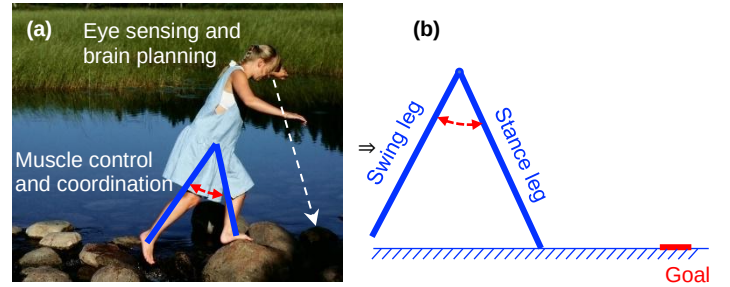


Figure 1. (a) Illustration of foot placement control problem. (b) Foot placement simplified model.

The stance leg has mass m_1 and length l_1 respectively. The angle the stance leg forms with the ground is θ_1 . Similarly, denote the mass and length of the swing leg as m_2 and l_2 , as well as θ_2 as the angle it creates with the vertical orientation. The modeling of the simplified foot placement model is shown in Figure 2.

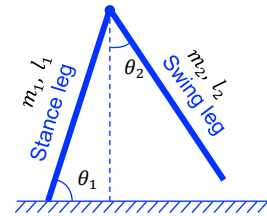


Figure 2. System dynamic modeling of the foot placement model.

The model under consideration can be categorized as a double-pendulum system with 2DOF. Here, the stance leg (or support leg) is represented by an inverted pendulum, while the swing leg behaves as a typical (or regular) pendulum. The dynamic model can be derived by Lagrangian mechanics approach as follows. Due to the center of gravity position measured by the "Suspension method" has a certain error and in order to simplify the experiment, in later experiments and simulations, the distance between the center of gravity and the

endpoint, that is l'_1 and l'_2 , are assumed to be consistent with the length l_{i1} and l_{i2} obtained by fitting the moment of inertia. The potential energy in the system is:

$$V = m_1 g l'_1 \sin \theta_1 + m_2 g (l_1 \sin \theta_1 - l'_2 \cos \theta_2) \quad (1)$$

The kinetic energy is:

$$T = \frac{1}{2} J_1 \dot{\theta}_1^2 + \frac{1}{2} J_2 (\dot{\theta}_2)^2 + \frac{1}{2} m_2 \left((\dot{\theta}_1 l_1 \sin \theta_1 - \dot{\theta}_2 l'_2 \cos \theta_2)^2 + (\dot{\theta}_1 l_1 \cos \theta_1 + \dot{\theta}_2 l'_2 \sin \theta_2)^2 \right) \quad (2)$$

Hence the Lagrangian is:

$$L = T - V = \frac{1}{2} J_1 \dot{\theta}_1^2 + \frac{1}{2} J_2 (\dot{\theta}_2)^2 + \frac{1}{2} m_2 \left((\dot{\theta}_1 l_1)^2 + (\dot{\theta}_2 l'_2)^2 + 2 \dot{\theta}_1 \dot{\theta}_2 l'_2 \sin(\theta_2 - \theta_1) \right) - m_1 g l'_1 \sin \theta_1 - m_2 g (l_1 \sin \theta_1 - l'_2 \cos \theta_2) \quad (3)$$

By applying $\frac{\partial(\frac{\partial L}{\partial \dot{\theta}_i})}{\partial t} - \frac{\partial L}{\partial \theta_i} = 0$, the equation of motion can be written in Matrix form as:

$$\mathbf{M}(\theta) \ddot{\theta} + \mathbf{C}(\theta) \dot{\theta} + \mathbf{G}(\theta) = \mathbf{B} \mathbf{u} \quad (4)$$

With

$$\begin{aligned} \mathbf{M}(\theta) \ddot{\theta} + \mathbf{C}(\theta) \dot{\theta} + \mathbf{G}(\theta) &= \mathbf{B} \mathbf{u} \quad (5) \\ \mathbf{M}(\theta) &= \begin{bmatrix} J_1 + m_2 l_1^2 & m_2 l_1 l'_2 \sin(\theta_2 - \theta_1) \\ m_2 l_1 l'_2 \sin(\theta_2 - \theta_1) & J_2 + m_2 (l'_2)^2 \end{bmatrix}, \\ \mathbf{C}(\theta, \dot{\theta}) &= \begin{bmatrix} m_2 l_1 \dot{\theta}_2 l'_2 \cos(\theta_2 - \theta_1) & m_2 l_1 l'_2 \cos(\theta_2 - \theta_1) (\dot{\theta}_2 - \dot{\theta}_1) \\ m_2 l_1 l'_2 \cos(\theta_2 - \theta_1) (\dot{\theta}_2 - \dot{\theta}_1) & -m_2 l_1 \dot{\theta}_1 l'_2 \cos(\theta_2 - \theta_1) \end{bmatrix}, \\ \mathbf{G}(\theta) &= \begin{bmatrix} m_1 g l'_1 \cos \theta_1 + m_2 g l_1 \cos \theta_1 \\ m_2 g l'_2 \sin \theta_2 \end{bmatrix}, \\ \mathbf{B} &= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \mathbf{u} = \begin{bmatrix} -\tau \\ \tau \end{bmatrix} \end{aligned}$$

where τ is the torque exerted by the hip and its positive direction is counterclockwise.

The reinforcement learning problem is formulated as follows. The dynamic system used, shown in [Figure 2](#), has four states, i.e., $\theta_1, \dot{\theta}_1, \theta_2, \dot{\theta}_2$. The time step length for the model simulation is 0.01 seconds. The initial state is any random value in the range of $\left[\frac{\pi}{3}, \frac{2\pi}{3}\right]$ rad for θ_1 and $\left[-\frac{\pi}{3}, \frac{\pi}{3}\right]$ rad for θ_2 . The initial

state range of $\dot{\theta}_1$ and $\dot{\theta}_2$ is in $[-3.5, 3.5] \frac{\text{rad}}{\text{s}}$. This is chosen to ensure a “reasonable” initial state. There is one actuation input to the system τ , which is the torque between the two legs provided by a hip actuator. The torque is discrete and there are three possible values: -5 Nm, 0 Nm, and 5 Nm, which represents negative direction drive, no drive, and positive direction drive. Since the rotational velocity change is integration of torque over time. The same velocity change can be achieved by exerting a small torque over a long period of time, or by exerting a large torque over a short period of time. As a result, this process has 2DOF. This results in a redundancy. For example, the same velocity change can be achieved by exerting a small torque over a long period of time, or by exerting a large torque over a short period of time. Hence the optimal control strategy is not unique. This results in a redundancy. The discrete driving torque removes this redundancy by fixing the torque in one drive direction. The system is trained to start from any given initial state and achieve one of two goal states as shown in the example in [Figure 3](#). These two goals are chosen to represent two steppingstones in real life and the walking robot has to achieve either one of these two goals to stay balanced.

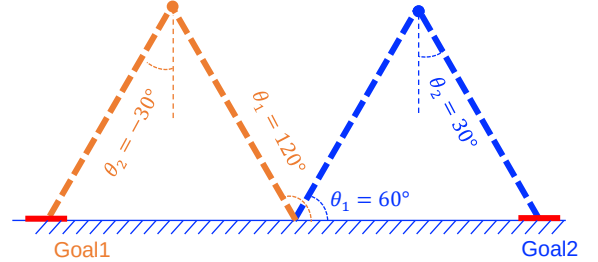


Figure 3. Illustrate of the control goals. Goal 1 (in yellow) with $\theta_1 = \frac{2\pi}{3}$ and $\theta_2 = -\frac{\pi}{6}$; Goal 2 (in blue) with $\theta_1 = \frac{\pi}{3}$ and $\theta_2 = \frac{\pi}{6}$.

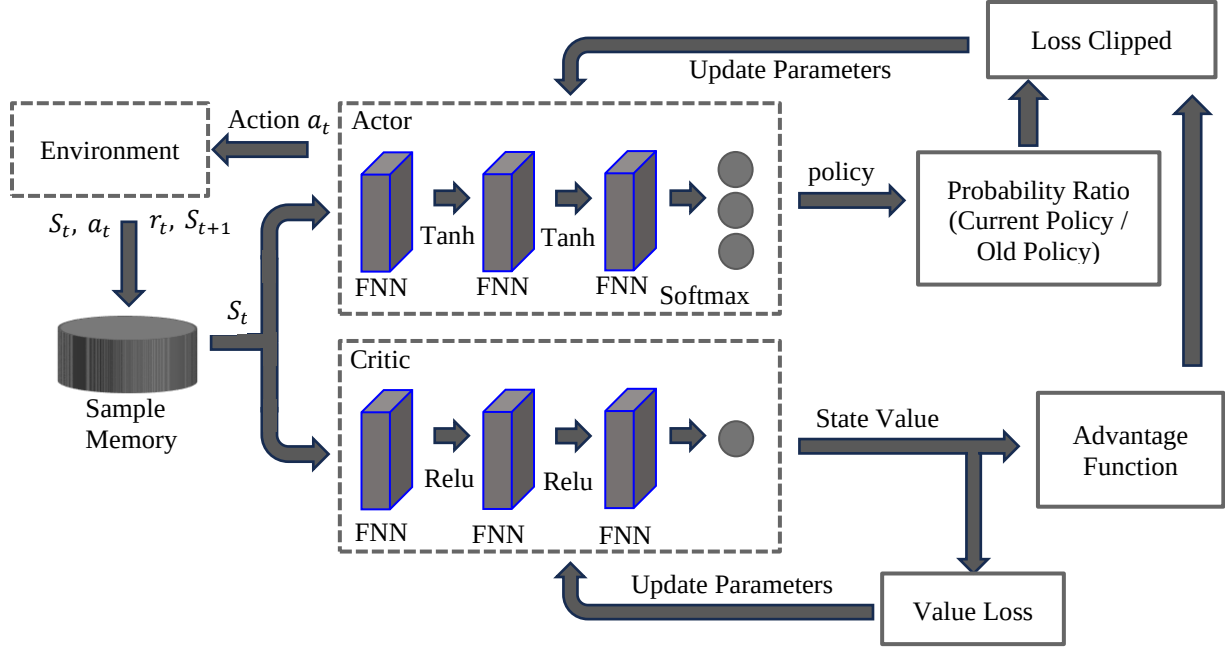


Figure 4. PPO algorithm For Foot Placement.

B. Deep Reinforcement Learning For Foot Placement

To achieve foot placement in the full state domain, an end-to-end deep reinforcement learning algorithm is designed, and the detail is shown in [Figure 4](#). For Critic, the data dimension received by the input layer is 4, and the output dimensions of the three linear layers are 1024, 512, and 1 respectively. For Actor, the data dimension received by the input layer is 4, and the output dimensions of the three linear layers are 1024, 512, and 3 respectively.

C. Target position detection and real-time state input

External visual analysis modules are assumed to exist, which can pass the next landing position to the model as target position 1 and target position 2. At the same time, there are encoders at the feet of the standing legs and the hips of the robot that can transmit the real-time angles of the legs to the model.

D. Reward Function

For a considerable portion of walking situations, failure to place the foot correctly at the target position means high risk, such as crossing a river by stepping on stones. Therefore, our reward is a sparse reward, that is, only when the foot is successfully placed near the target position S_{goal} whose allowed error is 5 degrees can it be rewarded, and a negative reward is given when the robot falls ($\theta_1 < 0$ or $\theta_1 > \pi$) or the leg structure is damaged ($\theta_2 < -\frac{\pi}{2}$ or $\theta_2 > \frac{\pi}{2}$). At the same time, in order to reduce excess energy consumption, a negative reward $\alpha\tau$ will be given every time the robot performs an action that outputs τ . Each time step in which an action is performed will

detect whether a reward r_t is obtained, which is formalized as follows:

$$r_t = \begin{cases} R_{\text{goal}}, & \text{reach } S_{\text{goal}} \\ -R_{\text{goal}}, & \text{fall down or break the leg} \\ -\alpha\tau, & \text{take action} \\ 0, & \text{elsewhere} \end{cases} \quad (5)$$

E. Passive control training

In order to achieve passive dynamics, the PPO-based reinforcement training is carried out based on -5 Nm / 0 Nm in a control sequence. This ensures that in a control sequence, no positive and negative torque drive show up one after another. In this way, the system is passive. The entire action map, *i.e.*, driving torque as a function of the four state spaces, $\tau = f(\theta_1, \dot{\theta}_1, \theta_2, \dot{\theta}_2)$ is obtained under this action torque setup.

However, since the system is driven only in one direction or no driving force at all, the initial states that have the potential to overshoot the target position cannot achieve the target position. Because any driven force that has the same direction as the overshoot direction will make the system overshoot even more, which make it impossible to achieve to the target position. In order to find out if the initial state can converge to the target position, each initial position is checked to figure out whether it can converge to the target position. If it cannot converge, it is marked as “not converged” in the action map and is denoted as dark blue.

A small part of the action map is illustrated in [Figure 5 \(a\)](#). This plot is done by setting $\theta_1 = 80^\circ$, and $\theta_2 = -20^\circ$ and then show the action map as a function of $\dot{\theta}_1, \dot{\theta}_2$. In this figure, yellow area shows that at this angle, part of the states (mostly with positive $\dot{\theta}_1$) can achieve the target position without any

torque control at this state. It should be noted that $\tau = f(\theta_1, \dot{\theta}_1, \theta_2, \dot{\theta}_2)$ is a continuous function but this figure is plotted with resolution of 30×30 resolution.

However, the same figure reveals that there is a large area that cannot converge to the target, which is shown by dark blue color. This problem is solved by adding another training sequence with two actions to choose from: 5 Nm / 0 Nm. As shown in Figure 5 (b), the training with 5 Nm / 0 Nm result in a control map that reaches converge for most of low and negative $\dot{\theta}_1$ areas. The system cannot reach the target state for most of the $\dot{\theta}_1 > \frac{\pi}{3}$ regions. This is reasonable because if the system at that region is left uncontrolled, it will undershoot the target position. As a result, a positive torque will make it even more undershoot and cannot reach the target position.

The combined action map is synthesized as follows. Two action maps with -5 Nm / 0 Nm, 5 Nm / 0 Nm action choices are trained and compared. For the one state, the action map will be -5 Nm or 5 Nm if any of those two shows up. According to the result, -5 Nm and 5 Nm will not show up for the same given state. The state point will be considered as not converged only if both action map consider it as not converged at the same time. Otherwise (i.e., the action is not -5 Nm, 5 Nm, or not converged), the action will be 0. The combined action map is shown in Figure 5 (c) and is used in the analysis later in this manuscript. This method is referred to as the passive method, with respect to the active method that have all -5 Nm / 0 Nm / 5 Nm torque choices.

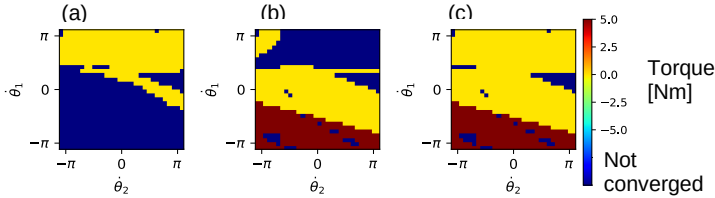


Figure 5 Action map obtained with training. (a) Action map obtained with -5 Nm / 0 Nm. (b) Action map obtained with 5 Nm / 0 Nm. (c) Action map obtained with the combination of controlling strategy in (a) and (b).

IV. SIMULATION RESULTS AND ANALYSIS

A. Simulation validation

Based on the action map obtained, simulation is done with initial position as $[\theta_1, \dot{\theta}_1, \theta_2, \dot{\theta}_2] = [67^\circ, 200^\circ/s, 60^\circ, 200^\circ/s]$. As soon as the simulation starts, the action torque is on according to the action map $\tau = f(\theta_1, \dot{\theta}_1, \theta_2, \dot{\theta}_2)$. It can be observed from Figure 6 (a) that with the actuation from the action map, the two legs eventually reached the target position at $\theta_1 = 120^\circ, \theta_2 = -60^\circ$. The detailed angles, angular velocities, and the action torque is shown in Figure 6 (b), (c), (d). More experiments are done with different initial conditions and the action map. It is shown that for all the converge initial conditions (indicated as light blue, yellow, and red colors in

Figure , except the dark blue color) can converge to the target position within an acceptable error of 5° .

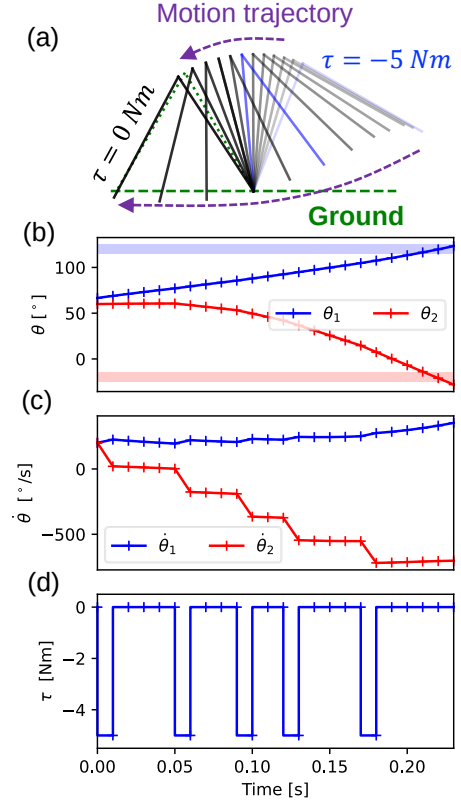


Figure 6. Simulation results. (a) The motion trajectory. The motion direction is illustrated as the purple line. The positive of all angle and torque is defined as the same as the direction of counterclockwise. The color transparency decreases as a function of time. (b) Motion angle in time domain; (c) Motion angular velocity in time domain. (d) Action torque in time domain. The light blue and light red region are the goal position for the stance leg and swing leg respectively.

B. Comparison between passive control and active control

In Sec. 3, 5 Nm / 0 Nm, -5 Nm / 0 Nm controlling torques are used in separate control sequences to ensure that positive and negative action torque will not show up in one control sequence, thus ensuring the system control is passive. In order to find out the influence of this passive setup, the same training and simulations are performed with three action torques to choose from: 5 Nm, 0 Nm, and -5 Nm. The action mappings is obtained and shown in Figure 16 and compared with the action map obtained by the passive method (shown in Figure 15). It is found that the overall distribution of the positive torque, negative torque, zero torque areas, and the not converged areas are similar. In order to understand the proportions between each area more intuitively, the four state spaces are divided into 50 parts respectively, and used the middle point of the state space as the starting point to conduct simulation tests, and obtained the result as shown in Figure 7.

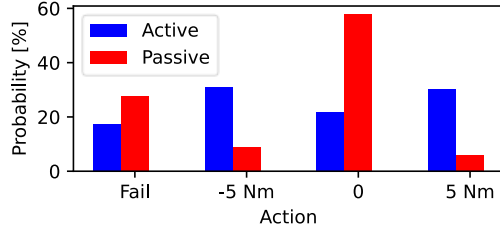


Figure 7. The proportion of the negative action, no action, positive action, and not converge cases in the entire action map.

Close examination reveals the following differences:

For almost all the “not converge” areas in the active method remain “not converge” in the passive method. There is a considerable area that does not converge in the passive method but converges in the active method. This indicates that being passive will decrease the system converge area. However, this phenomenon is not significant. As shown in Figure , the not converge” area increases from 17% with the active method to 27% with the passive method.

The no torque action increases from 22% with the active method to 58% with the passive method. This indicates that with the passive method, a majority of the action map has no torque at all. Meanwhile, with the increase of no torque action, the -5 Nm and 5 Nm torque action decrease from 31% and 30% to 9% and 6%, respectively. This means the system can be efficient since the motor is off for the no torque action and less of the time is with -5 or 5 Nm torque drive.

In order to better understand the difference between the passive control and active control, simulations with the same initial condition but with different action maps obtained by the passive and active method, are formed and shown in Figure 8. It's evident from the results that both systems successfully attain the target state. However, their motor active time is different. For the active control, the motor is active for 40% of the time in the control sequence, with 10% as negative torque output and 30% as positive torque output. It is obvious that positive and negative torque shows up in a controlling sequence. For the passive control case, only 10% of the whole control sequence is active but the system eventually reaches the same goal position. The passive control system necessitates shorter motor output torque time, making it a more energy-efficient control method.

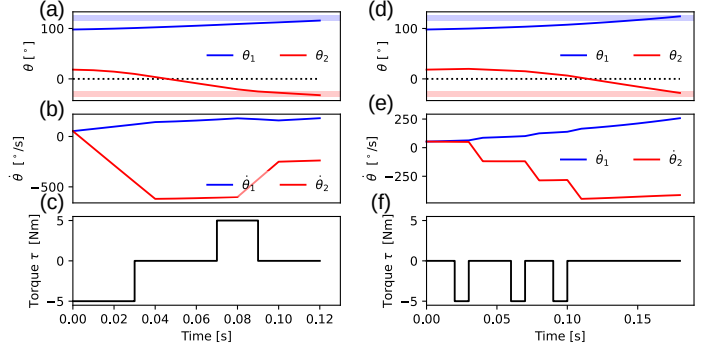


Figure 8. Simulation comparison. (a)(b)(c) The angle, angular velocity, and action torque of the active control method; (d)(e)(f) The angle, angular velocity, and action torque of the passive control method.

C. Reaction to disturbances

Walking robots face disturbances in real application scenarios and maintaining balance is a challenge for walking robots. The failures in the DARPA robotics challenge 2015 is a good example to showcase the balancing problem. Since the entire action map is obtained by training and reinforcement-learning-based control is one kind of optimal control, the robot can use the action map to get to the target position whichever the initial position is. The robot should be robust to disturbances. In order to demonstrate this, a disturbance is introduced to the control sequence and is shown in Figure 9. In this control sequence, the robot starts from some random initial position and tries to reach to the goal position from 0 to 0.28 seconds. The control sequence is successful, and the robot reaches close to its target. Suddenly, a disturbance happens in 0.28 s, and the robot is pushed to a random position with different angle and angular velocities for both θ_1 and θ_2 . However, the same control algorithm trained by reinforcement learning can still cause the robot to converge to the goal position, in this case, the goal position is different from the original goal position. This showcases the robustness of the control algorithm in disturbance.

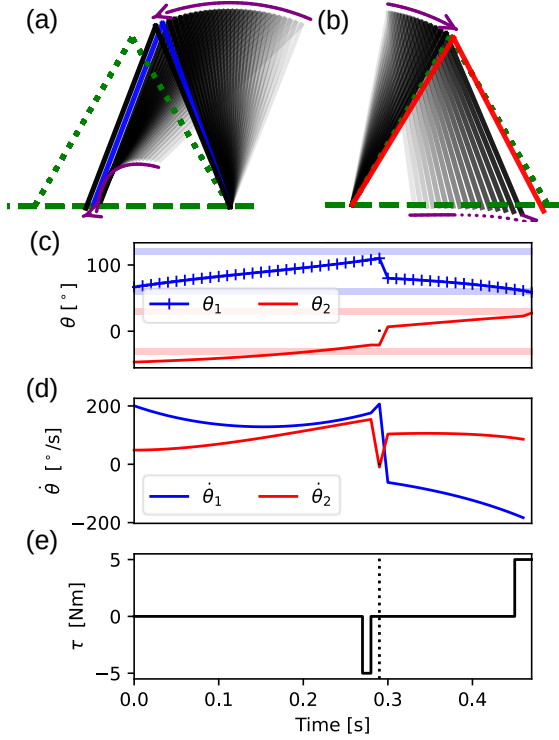


Figure 9. Robustness Test - State change. (a) the motion sequence before the disturbance. (b) The motion sequence after the disturbance. (c) the angles in the control sequence, including before and after the disturbance. The disturbance happens at 0.28 s and denoted as black dot lines. (d) the angular velocity in the control sequence. (e) the action torque in the control sequence.

D. Reaction to change of goal position

Human walking is not only robust but can also deal with complex situations. For example, the girl in [Figure 1](#) suddenly realizes the goal position she chooses is slippery. However, by large chance, she can still manage to control her leg to land on another target spot nearby. Similar control can be achieved by this reinforcement-learning-based robot. In [Figure 10](#), the simulation depicts a scenario where a person needs to change the goal position while walking. This is done by changing the action during the control sequence. For example, initially the system is controlled by an action map that is trained with a goal position as $[\theta_1, \theta_2] = [110^\circ, -20^\circ]$. A new action map with a new goal position $[\theta_1, \theta_2] = [130^\circ, -40^\circ]$ was trained before the control task. When the goal position suddenly changes, the new control sequence is generated according to the new action map. As observed in [Figure 10 \(e\)](#), the robot is about to reach the goal position $[\theta_1, \theta_2] = [110^\circ, -20^\circ]$ at 0.07 seconds. Suddenly, the new goal position changes, and the new control algorithm managed to get to the new goal position in the end.

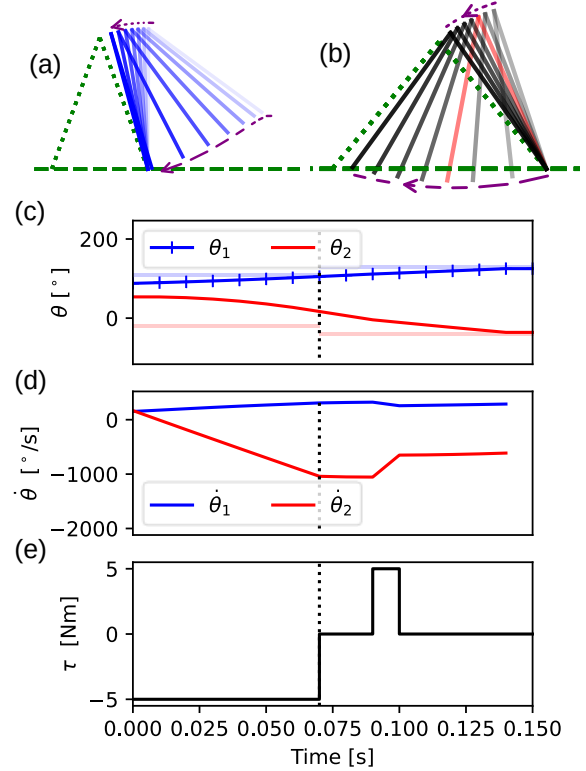


Figure 10 Robustness Test - Target change. (a) The motion sequence before the change of the target position. (b) the motion sequence after the change of the target position. The initial target position is $[\theta_1, \theta_2] = [110^\circ, -20^\circ]$ and suddenly changed to $[\theta_1, \theta_2] = [130^\circ, -40^\circ]$ in 0.07 s. (c) (d) (e) are the angle, angular velocity, and control torque respectively.

V. DESIGN, PROTOTYPE AND EXPERIMENTAL VALIDATION

In order to validate the modeling and simulation result, a prototype is built, and an experiment was carried out. A computer-aided design (CAD) sketch of the passive dynamic bipedal robot is conceived and shown in [Figure 11](#). The stance leg the bipedal robot is affixed to a base on one end and a hip motor is mounted on the other end. The stance leg is constructed using wooden boards, bearings, and photoelectric sensors. The photoelectric sensor's function is to measure the angle, denoted by θ_1 , between stance leg and the ground. In the robot's hip joint, a motor with a magnetic encoder is installed to measure θ_2 , i.e., the angle formed between the swing leg and the vertical direction. The motor in the robot's hip joint is adjusted to output a constant torque throughout the control phase by using the torque mode of the motor and setting the parameters of the torque mode. The motor is a Field-Oriented-Control-based brushless DC motor.

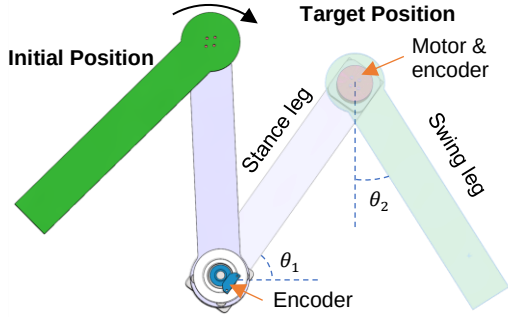


Figure 11. Physical design figure.

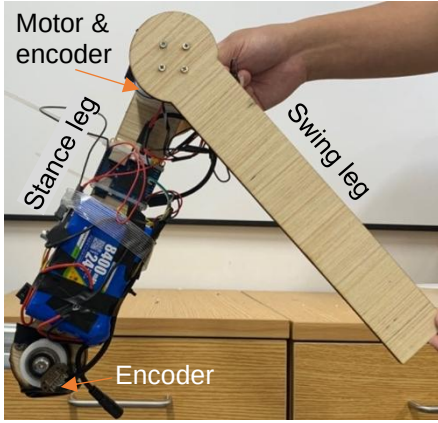


Figure 12. Prototype and experiment setup.

Figure 12 captures the assembled physical apparatus. In an effort to lower the influence of extraneous factors on the experiment, all sensors are securely fastened to the mechanical legs. The leg mass, length are measured, while the system moment of inertia and the torque generated by the motor are obtained by experiments. The motor torque is obtained by accelerating a rotation mass with its rotation inertia already known, and the result is shown in Figure 18. Through a free vibration test, the rotation inertial of the stance leg and swing leg are both obtained. The system properties are summarized in Table I.

Table I. Experiment setup system properties.

Parameter	Property	Value	Unit
Stance leg	mass	1412.2	<i>g</i>
Stance leg	length	330	<i>mm</i>
Swing leg	mass	83.9	<i>g</i>
Swing leg	length	330	<i>mm</i>
Battery	mass	980	<i>g</i>
Motor	mass	93.6	<i>g</i>
Encoder 1	pulse per revolution	65536	<i>cycle</i>
Encoder 2	pulse per revolution	200	<i>cycle</i>

Based on our prototype, a comprehensive series of experiments has been conducted, the outcome of which aligns well with our simulation results. It is validated that the swing leg can accurately reach the targeted position. With the system exhibiting only -5 Nm / 0 Nm or 5 Nm / 0 Nm, the control can be considered as passive. Having one motor and two joints lends the system to being underactuated. In theoretical terms, providing the initial state is reasonably set - that is, not in a stalled condition - the system should converge to the target position. Figure 13 offers one example of experimental data and simulation data. A notable congruence between the experimental outcomes and the simulated predictions can be distinctly discerned.

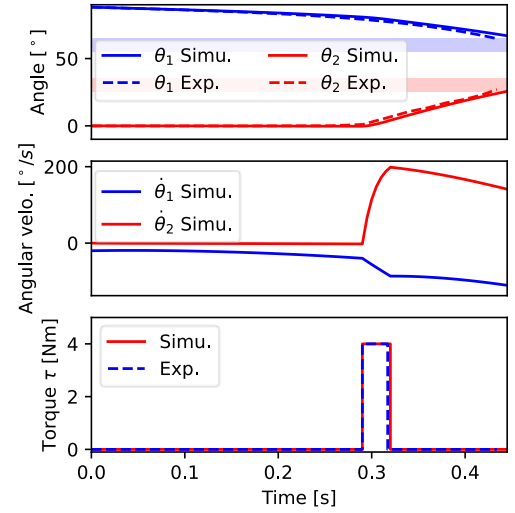


Figure 13. Compare modeling results with experimental results.

VI. CONCLUSION AND FUTURE WORK

In summary, we have successfully designed, trained, simulated, and validated a foot placement robot that impresses with its efficiency, robustness, and seemingly effortless operation. This particular experiment demonstrates the impressive capability of the control algorithm, which efficiently ensures posture recovery while operating under conditions of minimal energy consumption. At the same time, it expands the robot's robustness to a significant degree. The results shed light on the potential and prospect of fusing reinforcement learning techniques with passive dynamics.

Despite the promising results, certain limitations in our experimental procedures necessitates mentioning, such as the scuff feet problem. Future research should expand this idea and make it applicable to human walking foot placement control.

The elucidation of these shortcomings lays the groundwork for future research directions and potential improvements in the design of experiments, thereby heralding a more nuanced understanding of the complex interplay between reinforcement learning and passive dynamics in robotics. Our next stage will

be to advance the structural design of the robot. Building on our current progress, we plan to incorporate ankles into the robot's design, increasing its adaptability to more complex environmental conditions, leading to more flexible and robust control.

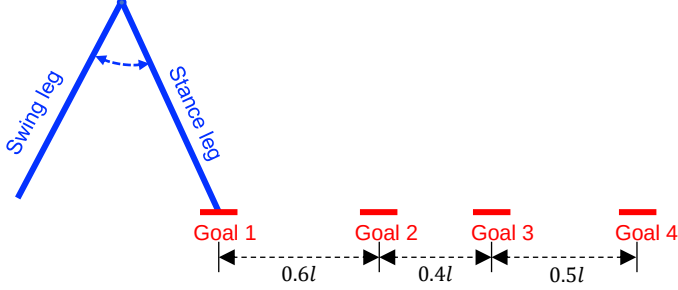
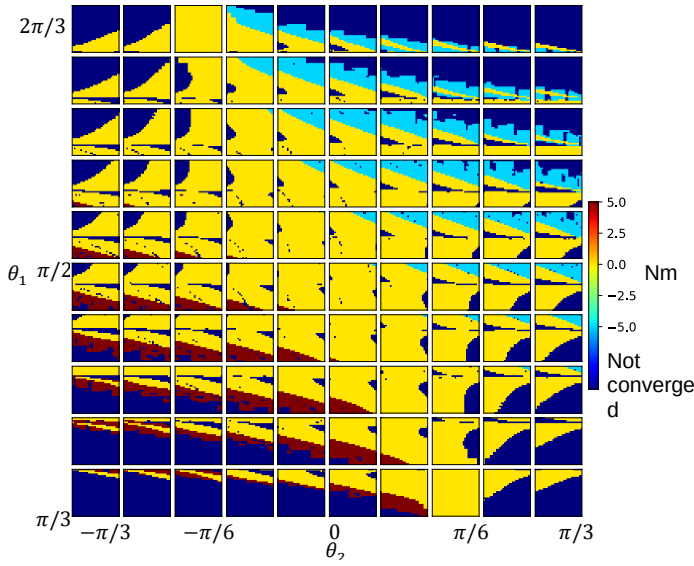


Figure 14. Future plan. A simplified bipedal walking robot based on passive dynamic and reinforcement learning. The robot can step at a set of pre-set goals precisely and achieve walking that is close to humans.



APPENDIX

Figure 15. Action map of passive control.

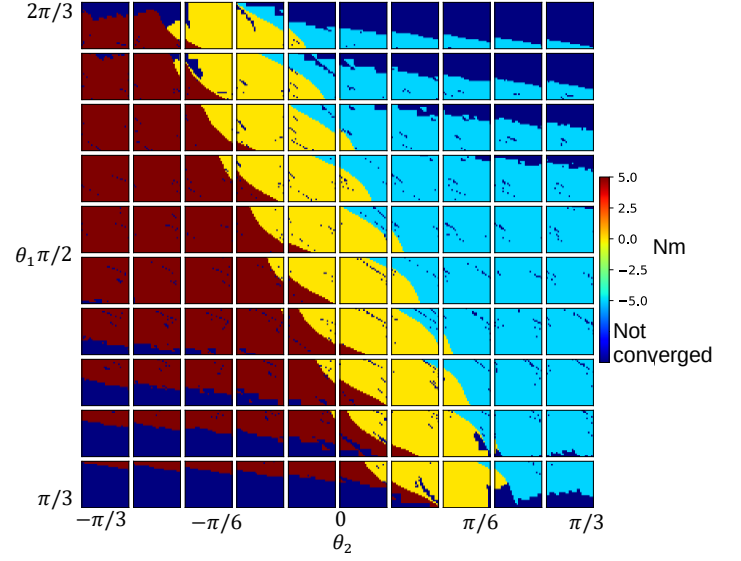


Figure 16. Action map of active control.

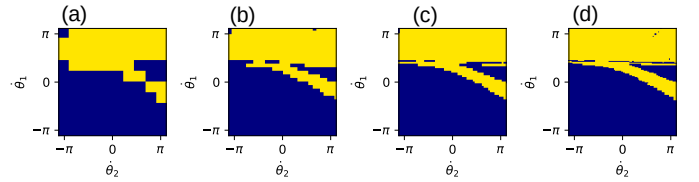


Figure 17. Action map resolution test. The same action map (as shown in Figure 15) with different resolution. (a) 10×10 ; (b) 30×30 ; (c) 50×50 ; (d) 100×100 .

After training $\tau = f(\theta_1, \dot{\theta}_1, \theta_2, \dot{\theta}_2)$ becomes a continuous function, however, for deployment and illustration purpose, digitization has to be done. In order to test the influence of resolution. Upon scrutiny of Figure 17, it becomes apparent that as N increases progressively, the resolution of the action map sharpens. However, that is at a cost of a more time-consuming calculation process and larger look-up table during deployment. Therefore, in the final experiment, the action map is discretized into a $50 \times 50 \times 50 \times 50$ table to balance resolution and load on the experimental equipment.

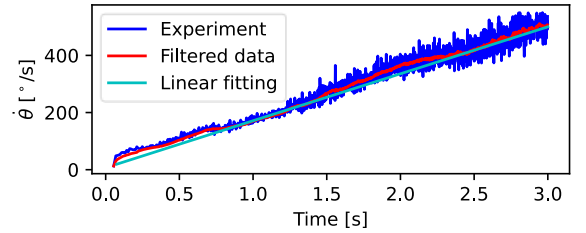


Figure 18. Torque test experiment. A constant torque is applied to the motor and a flywheel is connected to the output shaft. The rotation speed $\dot{\theta}$ is recorded and analyzed.

To ascertain the moment of inertia subsequent to parts addition, and to assess the damping properties of the bearings and motors, we secured one appendage of the legged robot, allowing it to dangle freely. A predefined initial velocity was imparted, enabling the capture of data pertaining to the temporal evolution of angle and angular velocity. *Figure 19* delineates the correlation between the simulated motion curve and the experimentally derived data curve. The latter was grounded in the moment of inertia and kinematic equation post the incorporation of the damping coefficient for the legs.

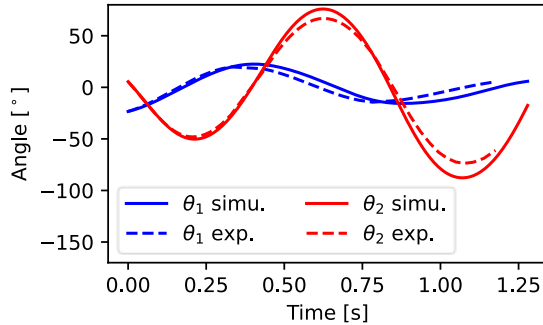


Figure 19. Pendulum get rotational inertia.

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